



# Solving Batched Initial Value Problems on GPUs

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## Initial Value Problems (IVPs)

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$$\begin{cases} \dot{\mathbf{y}} = f(t, \mathbf{y}; \mathbf{p}), \\ \mathbf{y}(t_0) = \mathbf{y}_0, \end{cases}$$

$\mathbf{y}(t)$ : state vector,  $\mathbf{p}$ : vector of fixed parameters. Goal: given many different  $\mathbf{y}_0, \mathbf{p}$ , compute  $\mathbf{y}(t_f)$  for each system.

- Important application: astrochemistry/combustion. We need to solve one IVP for each spatial grid cell, for each (outer loop) time step. This is a major computational bottleneck in these simulations; it is necessary to have surrogate/fast batched solver.

The two most simple time integrators are **forward** and **backward Euler**:

$$\mathbf{y}_{k+1} = \mathbf{y}_k + \Delta t \cdot f(t_k, \mathbf{y}_k; \mathbf{p}) \quad (\text{FE})$$

$$\mathbf{y}_{k+1} = \mathbf{y}_k + \Delta t \cdot f(t_k, \mathbf{y}_{k+1}; \mathbf{p}) \quad (\text{BE})$$

- Both extremely simple to implement, assuming the time step is fixed. (BE) requires a nonlinear equation solver. To parallelize: solve one ODE per thread.
- As  $\Delta t \rightarrow 0$ , the numerical solution approaches the true solution.
- These are examples of **first-order methods**: the error of the solution decreases at a linear rate.

## Adaptive Time-Stepping

In most applications, we would like to change the time step to match the dynamics. This is done by considering two time-steppers of different order and comparing the difference. For example: **RK23** (Bogacki, Shampine 1997):

$$k_1 = f(t_k, \mathbf{y}_k)$$

$$k_2 = f\left(t_k + \frac{\Delta t}{2}, \mathbf{y}_k + \frac{\Delta t}{2}k_1\right)$$

$$k_3 = f\left(t_k + \frac{3\Delta t}{4}, \mathbf{y}_k + \frac{3\Delta t}{4}k_2\right)$$

$$\mathbf{y}_{k+1}^{(3)} = \mathbf{y}_k + \frac{\Delta t}{9}(2k_1 + 3k_2 + 4k_3)$$

$$t_{k+1} = t_k + \Delta t$$

$$k_4 = f\left(t_{k+1}, \mathbf{y}_{k+1}^{(3)}\right)$$

$$e_{k+1} = \frac{\Delta t}{72}(-5k_1 + 6k_2 + 8k_3 - 9k_4).$$

This is a third-order method. The size of  $e_{k+1}$  determines the size of the next time step. For example:

$$\Delta t^{(k+1)} = 0.9 * \Delta t^{(k)} * \left( \frac{\epsilon_{\text{tol}}}{e^{k+1}} \right)^{0.2}.$$

**From this, a parallel computing challenge for GPUs arises.**

- If each ODE is solved on a single thread, some ODEs may reach  $t_f$  faster than others. Hence many threads will remain idle while the others finish.
- SIMT will break: we would like similar time steps to take place on the same block.
- Solution: cluster the initial conditions and parameters. Sort these before launching, so that each warp contains ODEs that are similar to each other. This maximizes synchronization inside thread blocks.

Step 1: Build a very simple batched  
IVP solver

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## The simplest case

- Explicit time-stepping (forward Euler), constant time step.
- Use Python **CuPy** interface, with **Raw Kernels** for solvers.
- To include custom RHS, embed text directly into the CUDA kernel (written in C) before compiling.
- Compare two different kernels:
  - `forward_euler_kernel`: on one thread, takes a single time step.
  - `forward_euler_kernel_multistep`: on one thread, runs a single IVP to completion.

Which is faster?



## Single Time Step vs. All at Once

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Include results here

